Benchmark Radar: A Living Database and Search Engine for AI Benchmarks and Evaluation

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Abstract

Benchmark researchers and developers of large language models (LLMs) and other AI systems need to find relevant evaluations, locate their benchmark datasets and code, and understand the settings behind reported scores. We present Benchmark Radar, a living database and search engine for retrieval and discovery of AI benchmarks, covering LLM evaluation, agentic and tool-use benchmarks, coding, reasoning, safety, and domain-specific evaluations. The system combines daily discovery of benchmark papers, repositories, datasets, and releases with a searchable benchmark catalog, mentions in model cards and technical reports, and score histories. It retains source identities and citations so readers can inspect candidate benchmarks and their evaluation evidence. Daily discovery draws on 37 sources: 13 direct connectors and 24 first-party research and engineering feeds. The catalog contains 1,283 source records drawn from 4 benchmark catalogs and 12,916 numeric observations on 790 records. We describe collection and retrieval, audit the full catalog, and examine benchmark saturation, adoption trends, and the limits of score comparisons. A worked example walks through a complete prior-art search, showing how to query the catalog and inspect benchmark evidence when designing a new evaluation. We release the web dashboard with a benchmark leaderboard, a Pareto frontier view of score against measured use, saturation and trend views, daily feeds, downloadable evidence, a command-line interface (CLI) for offline queries, and reproducible analysis.

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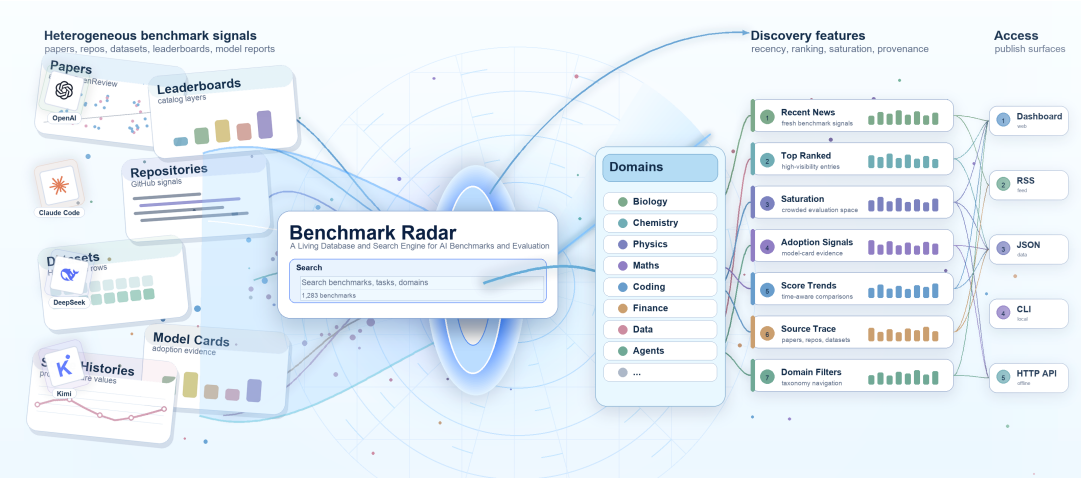


Figure 1: Benchmark Radar connects discovery, retrieval, and source inspection.

1 Introduction

The Transformer architecture established attention-based sequence modeling as a foundation for modern large language models (LLMs) [69]. GPT-6 Astra and Claude Fable 5.1 are designed for coding, research, and tasks that span multiple tools [3, 47]. Comparing these models requires benchmarks that expose failures and distinguish capability gains from changes in prompts, data splits, or evaluation procedures [21, 51, 54, 59].

Evaluation also matters beyond general-purpose chat. In recommender systems, Transformer-style sequence models and LLM-based methods support user, item, and preference modeling, with efficient training needed at production scale [29, 57, 66]. Foundation models are also used to model biological, geoscientific, and physical processes [13, 58, 68, 73, 81]. In social and political analysis, language models support text annotation and simulated human samples, while network methods quantify balance and polarization dynamics [4, 16, 35, 36, 46, 60]. Benchmark scores inform claims of progress across these scientific, industrial, and decision-support settings.

Benchmarks differ in the abilities they test. Broad evaluations of knowledge and reasoning include MMLU [21], GPQA [54], and Humanity's Last Exam [51]. Others isolate more specific capabilities: SWE-bench measures whether models can resolve real GitHub issues [27], LiveCodeBench tests code generation with date-based splits to reduce contamination from training data [26], and Terminal-Bench and Long-Horizon-Terminal-Bench test command-line agents on operational tasks [37, 41]. The Harbor framework ports more than 80 such benchmarks to a common execution contract so that arbitrary agents can be evaluated against them, and its Harbor-Index curates 82 difficult tasks drawn from 29 of those benchmarks [20, 61]. Domain-specific evaluations include FinanceBench for finance and economics [25], MATH for mathematics [22], SciBench for scientific problem solving [70], LAB-Bench for biology [34], and ChemBench for chemistry [42]. ESM-Bench tests agents' understanding of Earth system model physics and code [74]; ResearchClawBench evaluates end-to-end autonomous scientific research [78]; and ASI-Bench examines scientific exploration and execution with progressively less methodological guidance [82]. ScienceIDE converts scientific code repositories into executable, agent-learnable environments for training and evaluating scientific agents [15]. Benchmark construction is itself being automated: Benchmark Agent builds benchmarks end to end, from query analysis and subtask design to annotation and quality control [77]. Their tasks and scoring rules specify which model behaviors count as successful performance.

Finding a benchmark, its dataset or repository, and reports of its use requires searching paper servers, code hosting sites, dataset hubs, vendor releases, blogs, and benchmark catalogs. Existing resources such as LLM Stats, OpenCompass, and Artificial Analysis provide leaderboards, evaluation platforms, and model-analysis views [5, 39, 48]. Connecting newly released benchmarks to their task materials and later use in model reports can still require consulting several separate resources. Benchmark Radar addresses this gap by combining catalog retrieval with daily discovery, mentions in model reports, and scores with documented evaluation settings and sources.

Benchmark Radar gathers benchmark-related artifacts from public sources, groups observations with matching identifiers, and exposes a searchable index with source labels, dates, benchmark mentions, and scores (Figure 1). We present the search system, describe the collection and ranking methods needed to reproduce its outputs, and examine benchmark coverage, documentation and scored-model coverage across all catalog sources, source concentration, and gaps in evaluation settings.[1]

The system preserves one record per source benchmark, including entries without scores, dates, or citations. Reviewed identity links connect related records while retaining their separate measurements. We contribute this living search system, shared web and offline access to its evidence, and a repro-

[1] Source code: https://github.com/ktwu01/benchmark-radar.

ducible full-catalog census. Section 5 shows how a contributor used retrieval and source inspection to assemble prior art.

2 Related Work and Scope

Benchmark catalogs and evaluations. LLM Stats publishes benchmark descriptions and reported model results [39]. OpenCompass provides an evaluation platform and a benchmark registry with artifact metadata [48]. Artificial Analysis publishes model evaluations and their methodology [5]. Benchmark Radar uses records from these sources alongside model-report evidence. Benchmark Radar adds daily discovery, artifact histories, benchmark retrieval, and source inspection through shared web and offline access, helping readers investigate evaluations across the contributing catalogs.

Documenting evaluation evidence. Model cards and datasheets motivate documenting evaluation conditions and dataset characteristics [14, 43]. Benchmark Radar retains that information when the collected sources provide it, with citations for follow-up review. Benchmarks such as GPQA [54], SWE-bench [27], and SciBench [70] define different tasks. The census measures which evidence is available for inspection and which metadata still needs review.

Evaluating benchmarks themselves. Recent work examines benchmark saturation, item quality, and the interpretation of aggregate scores. In a systematic study of 60 language model benchmarks, Akhtar et al. [1] define saturation and report that nearly half exhibit it, with prevalence increasing with benchmark age. They associate resilience with expert curation rather than test-data privacy. Sample-level auditing of MMLU, ARC, WinoGrande, HellaSwag, and TruthfulQA identifies within-benchmark variation obscured by aggregate accuracy [62]. A reference-free judging framework assesses conversational-agent benchmarks on consistency, complexity, and policy coverage [33]. For safety benchmarks developed for larger models, rankings of smaller models vary with the treatment of ambiguous responses [56]. Inverse-density weighting reduces the influence of benchmark multiplicity on aggregate scores [38]. Gilda and Gilda [17] argue that evaluation scores should state their evidential scope and validity window, with conservative aggregation when component signals differ in reliability.

Benchmark Radar complements these methods by making benchmark records, available measurements, and provenance searchable. We used its command-line client to identify candidate papers for this section, supplemented the results with coauthor recommendations, and reviewed the source papers.

3 System and Methods

3.1 System Overview and Daily Discovery

Benchmark Radar adapts BuilderPulse's daily public-source collection approach [76]. Figure 2 separates the benchmark catalog from discovery history. Model reports include model cards, technical reports, system cards, and release posts; they contribute through the same record structure as benchmark registries.

Table 1 summarizes the four catalog sources and their primary uses in the v0.11.0 release.

Catalog source	Rows	Primary use
LLM Stats	687	Benchmark names and descriptions with source provenance.
OpenCompass Hub	461	Paper, repository, release, and dataset links with source-specific identity fields.
Artificial Analysis	25	Current commercial evaluation catalog entries.
Model reports	110	Benchmarks mentioned in model reports, including records without scores.
Total	1,283	All four sources, retaining each record's source label.

Table 1: Catalog sources in the v0.11.0 release. Rows count source-specific benchmark records, including records without scores.

3.2 Discovery Collection

Daily discovery observations describe mentions, releases, and updates found across public sources. An observation is one collected record; an artifact is a paper, repository, dataset, release, or page linked by exact identifiers. These objects differ from source-specific benchmark records. We retain their histories alongside the catalog without adding discovery observations to the benchmark total.

Each collection run searches a 48-hour window, records counts and errors by source, removes future-dated rows, and requires healthy core sources before publication. At the cutoff, arXiv [6], Hugging Face Hub [24], and GitHub Search [19] were the core sources. Across 13 direct connectors and 24 first-party feeds, additional routes cover scholarly indexes, dataset hosts, repository releases, and institutional feeds. Appendix D records their cutoff status.

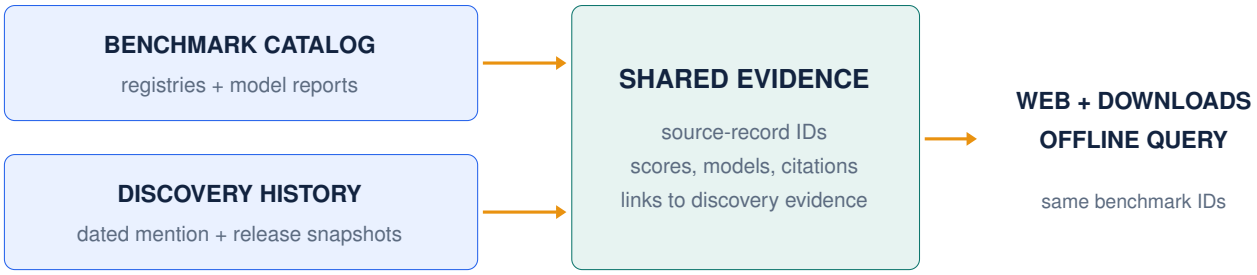


Figure 2: Two input paths support publication: dated discovery snapshots preserve mention and release evidence, while benchmark registries and model reports populate the shared catalog. Clients use the catalog's stable source-record IDs.

Table 2 describes the interfaces. Search and exports retain the full catalog; display filters affect only the selected view.

Interface	Reader question	Evidence and scope
Today	What appeared or changed?	Daily discovery observations, source health, and links.
Search	Which benchmarks match this task?	The complete catalog, including unscored records.
Leaderboard	Which evaluations combine lower scores with broader measured use?	Scored source records; distinct scored models or cited documents supply the selected count.
Saturation	What scores and settings were reported?	Benchmark browsing and complete score histories; search reaches every source and year.
Blog	Can I read and share a daily brief?	Daily briefs and an archive built from committed discovery snapshots.
CLI	Can I inspect the same evidence locally?	The offline query client uses the same benchmark IDs and stable JSON response format.

Table 2: Reader questions and evidence scope.

3.3 Preserve Records Before Comparing Measurements

A source record is one benchmark entry from one contributing source. We normalize names, identifiers, artifact links, score observations, model identities, and cited documents into common fields. A record remains in the catalog if a score, date, or citation is absent. Reviewed identity links connect related records while preserving their separate observations and counts.

Daily discovery contributes a different kind of evidence: collected mentions, releases, and updates. Exact identifiers such as DOIs, arXiv IDs, and repository URLs link those observations to artifacts. Discovery observations do not increase the benchmark catalog total. Appendix C records detailed collection settings and display filters; Appendix D records source health at the cutoff.

3.4 Retrieve Candidates with Their Evidence from the Dashboard or the CLI

The same benchmark IDs reach web search, detail pages, dataset exports, and offline clients. Lexical search uses BM25F, a field-weighted word-matching score [55], with bounded boosts for name and phrase matches. Each result exposes matched and missing query words, the fields they occur in, and the score components. Source membership does not change the ranking. A shared query service supplies the CLI and HTTP interfaces with the same response format and local data provenance.

The CLI keeps its own copy of the data. `benchmark-radar init` downloads the dataset archives and `benchmark-radar sync` updates them; a local manifest verifies each archive's SHA-256 checksum and records its schema version and provenance. Queries then run without network access, so a search can be repeated against a recorded dataset version. A search covers the normalized source records (`catalog`), the daily discovery snapshots (`radar`), or both (`all`). Filters restrict results by modality, openness, source, and the presence of a paper, repository, or dataset link. `benchmark-radar serve` answers the same queries on a local HTTP endpoint for programmatic use.

Commands that return results accept `--json`, which prints a versioned payload holding the ranking explanation above, the matching policy applied, and the version and build date of the local data. The same interface ships as an agent skill (`npx skills add ktwu01/benchmark-radar`) that states when a request calls for a benchmark query, which queries to run, and how to install the CLI and initialize

its data when they are absent. A reader and an agent then work from the same records and the same score components.

Candidate retrieval precedes suitability judgment. An analyst or agent can try focused query variants, inspect a record's tasks and score settings, and follow its citations. The interface preserves the evidence needed for that review. This paper evaluates catalog and measurement coverage; the contributor case illustrates usage without measuring retrieval accuracy or time saved.

3.5 Audit the Full Population

The census starts with every record in the rebuilt catalog index and reads its detail file. It applies no date, score, or interface filter. We count finite numeric score observations once by observation ID. Within each benchmark record, we count scored models by source model ID, preserving separately evaluated configurations, and cited documents by document ID. Repeated observations do not create additional models or documents. The global model registry uses its recorded identity links; per-benchmark model counts are not summed into a global total.

Eligibility depends on the measurement a calculation needs. A declared percentage unit, known score direction, and numeric values within 0–100 permit a percentage-scale summary. Rescaling a displayed value or reading an aggregator's declared maximum does not establish that unit. Matching scales also do not establish matching test versions, prompts, tools, attempts, or evaluators.

Date coverage counts valid recorded benchmark release dates. Score entries retain their own date basis, including model announcements and document publication. We do not substitute those dates for an evaluation date. Appendix F provides the software revision, input hashes, census script, and validation procedure.

4 Results

Appendix A provides the complete linked census; Appendix B details document, model, score, and date coverage.

4.1 Catalog Coverage and Task Materials

The rebuilt catalog contains 1,283 source records across 4 sources. We found 12,916 numeric score observations on 790 records, with 493 records lacking numeric scores (Table 3). Two sources can describe a related benchmark, and we retain both source records.

Source	Records	Scored	Unscored	Numeric scores
LLM Stats	687	679	8	5,544
OpenCompass Hub	461	0	461	0
Artificial Analysis	25	25	0	7,050
Model reports	110	86	24	322
Total	**1,283**	**790**	**493**	**12,916**

Table 3: Score coverage across the full catalog: 790 records have numeric observations and 493 do not. Scores count observations, not distinct models.

Of the 493 unscored records, 464 have at least one paper, repository, or dataset link. In the full catalog, 475 records link to papers, 506 to repositories, and 293 to datasets. A record can contain

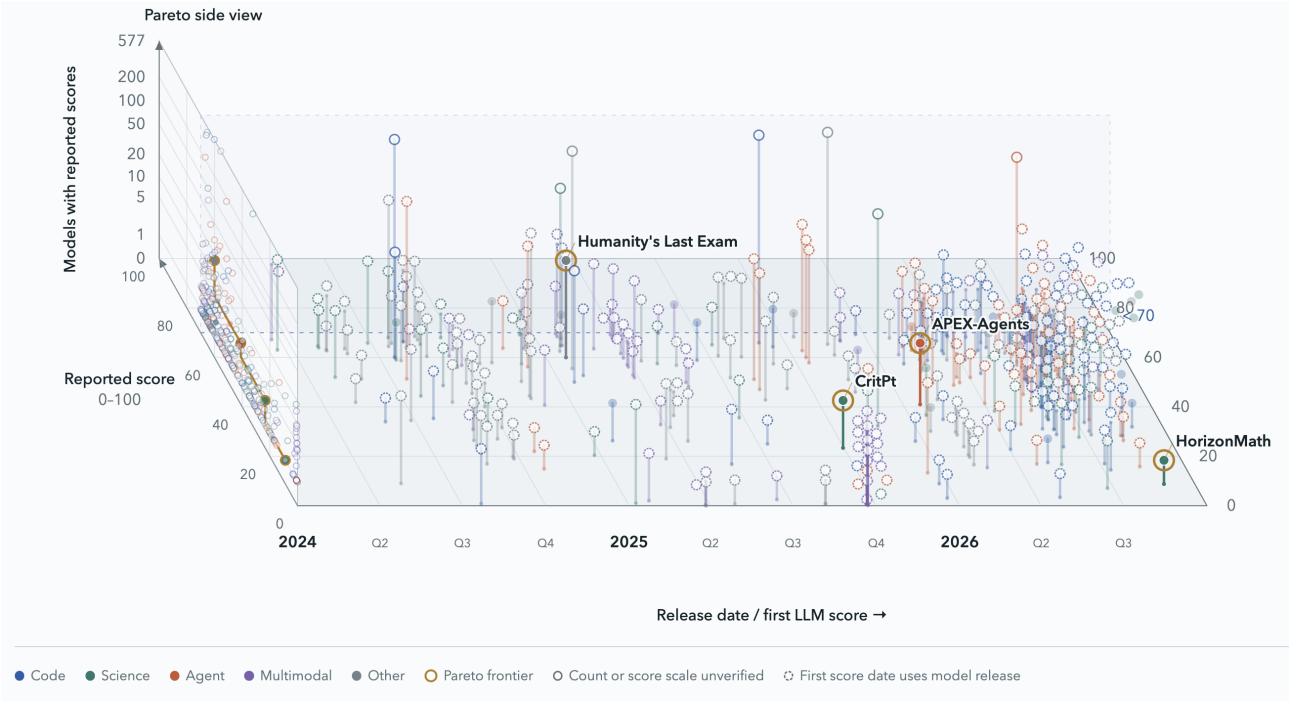


Figure 3: Benchmark Frontier on the Leaderboard page. The view combines reported scores, scored-model counts, and benchmark release dates or first-score date proxies. Gold rings mark the interface's Pareto candidates; hollow marks flag unverified scales or counts, and dotted outlines flag model-release date proxies. Click the image to explore the view.

more than one type of link. For example, the OpenCompass Hub record for A-Bench links a paper, code repository, and dataset despite having no archived score.

For scored records, Figure 3 lets readers browse by date, reported score, and number of scored models. Figure 4 shows the reported model scores within one source record.

4.2 Benchmark Taxonomy Across the Full Catalog

Most source records carry a capability label. We classify 1,279 of 1,283 records into 11 top-level domains and 63 sub-domains (Figure 5). Labels come from the publishers' own fields—OpenCompass Hub dimensions, LLM Stats categories, Artificial Analysis categories, and the model-report registry domains—so these 1,279 records trace to a named source field. The remaining 4 records are LLM Stats community rows whose crawl supplied no description, category, or modality; we record that reason rather than assigning a class from the title.

Interaction paradigm and input modality are recorded as *facets*: properties held beside a record's domain rather than inside it, so every record carries exactly one Level 1 class and, independently, any number of facet values. Facets therefore overlap each other and the Level 1 classes, and their counts do not sum to the population.

The separation matters because a benchmark that resolves repository issues is a coding benchmark run as an agent, not an agent benchmark. Of the 345 agentic records, 128 take Agentic & Tool Use as their Level 1 class and 117 sit under Coding & Software Engineering, with the rest spread across 6 further classes. A scheme with one axis has to choose, and choosing the domain hides those 117 records from any count of agentic evaluation.

Figure 6 puts each of the 615 records with a benchmark release date on its release year. The

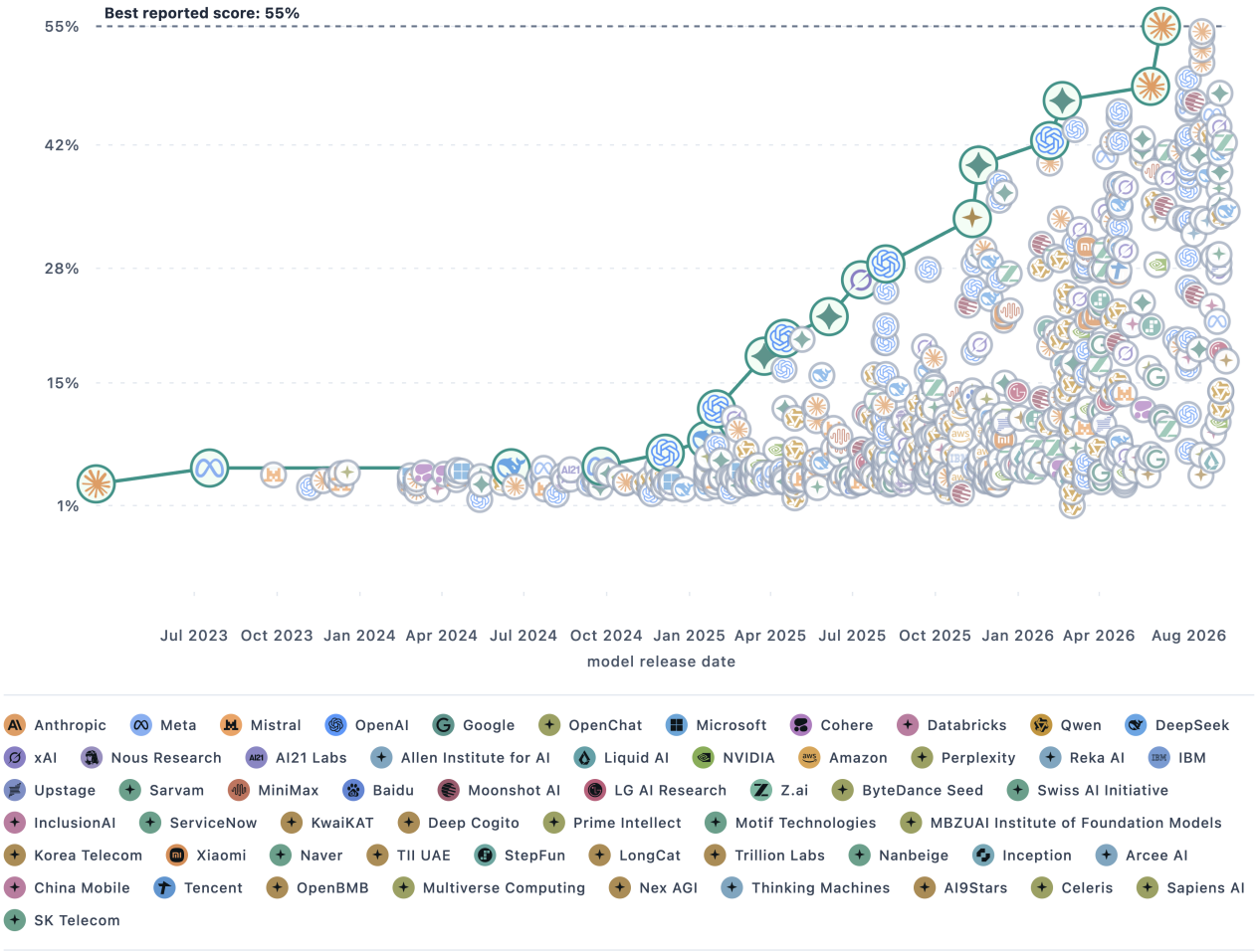


Figure 4: Inspecting Humanity's Last Exam in Benchmark Radar. This Artificial Analysis record contains 577 reported scores; the highest displayed score is 55.47. Marks identify model organizations, and the line connects successive best scores ordered by model release date. Click the image to open the interactive view.

668 records without a release date stay in a separate column with the same vertical scale. A large record count is not sufficient to report the share of a year. The OpenCompass Hub crawl stops at the discovery cutoff, so model reports supply most records in the last, incomplete year. A share for that year would show a change in the catalog, not in the field. Thus, the source mix of a reported year must also be near the pooled mix. Over the reported years, this mix is stable. If we reweight each year to a common source mix, the agentic share changes by no more than 1.1 percentage points. Thus, the increase in the agentic share does not come from the source mix.

4.3 Discovery Coverage Alongside the Catalog

The daily discovery collection contains 11,068 observations and 6,546 artifacts linked by exact identifiers across 46 snapshots. 4 snapshots are simulated historical backfills; their dates describe reconstructed collection windows. These discovery units remain separate from benchmark records.

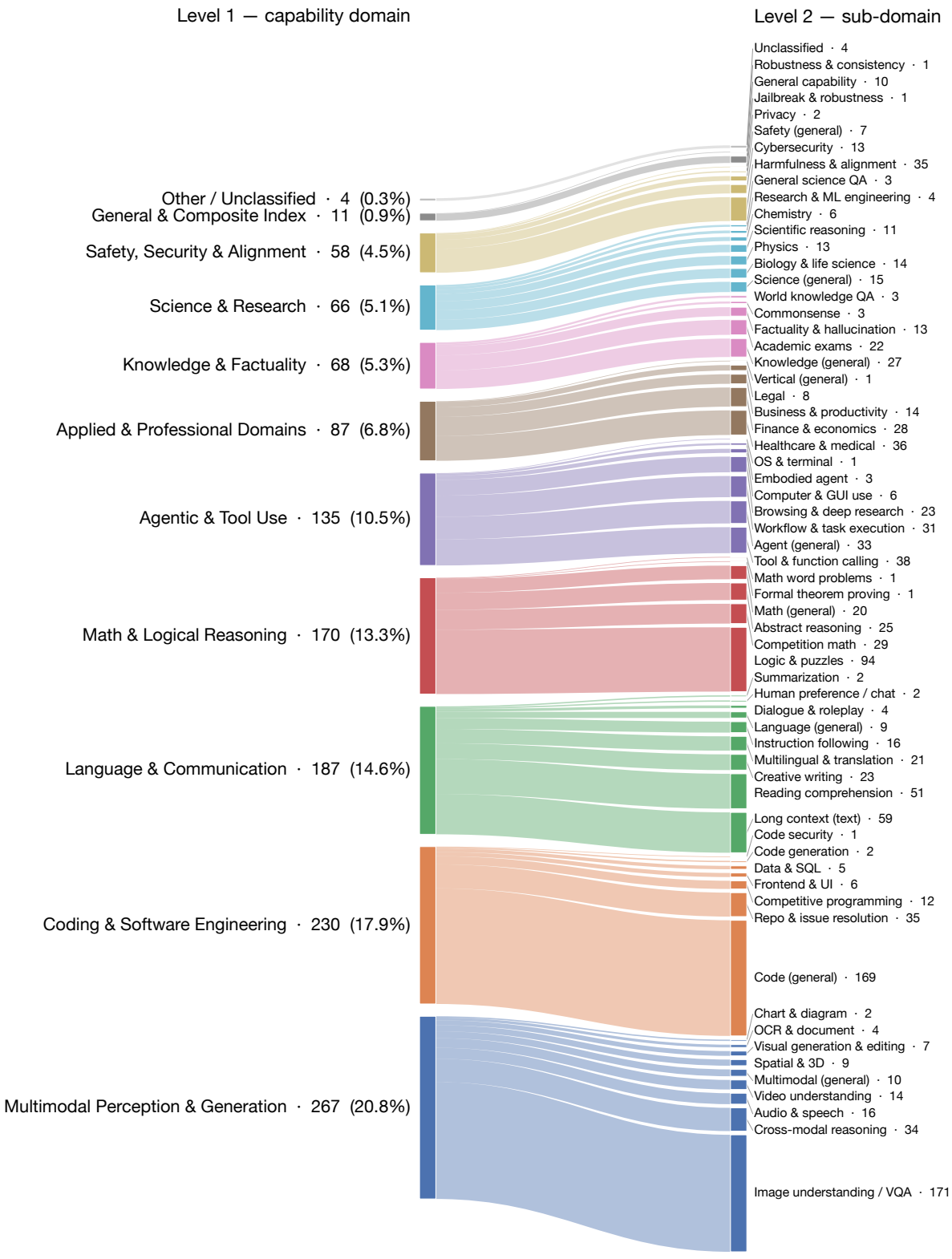


Figure 5: Capability taxonomy over the full catalog. Ribbon height is one unit per source record, so both columns carry all 1,283 records. Level 1 is one primary label per record; interaction paradigm and modality are recorded as separate facets and are not drawn here.

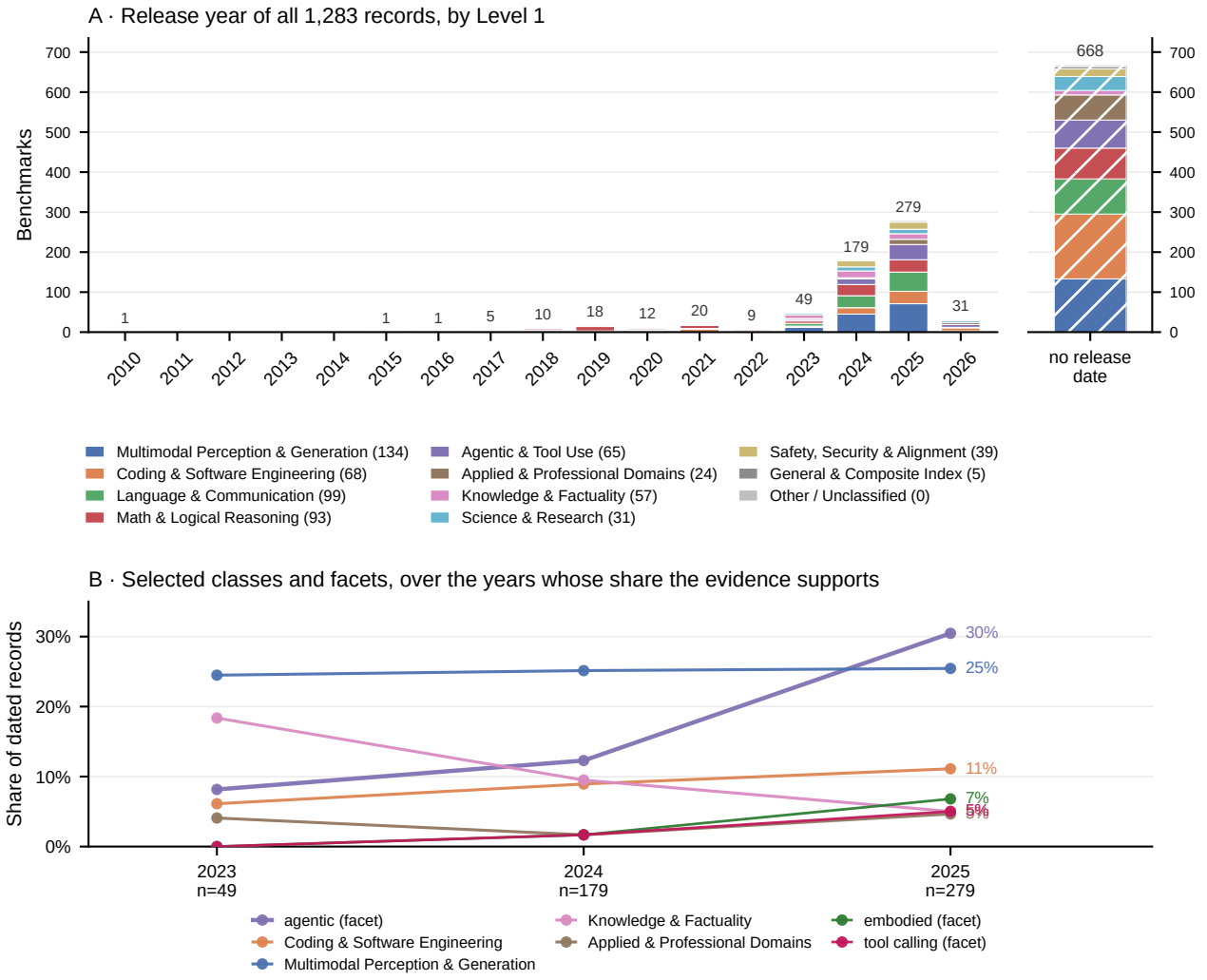


Figure 6: Release-year composition of the classified catalog. (A) All source records by release year and Level 1 domain. The year axis holds the 615 records with a release date. The 668 records without a release date are in the hatched column. This column uses the same vertical scale as the year columns. (B) Shares of selected domains and facets among the dated records of each year. Panel B shows only the years that have a minimum of 30 dated records and a source mix near the pooled mix. The other years have 108 dated records, and Panel A shows these records. Series marked *facet* are properties that cross the Level 1 classes. Thus, the lines in Panel B overlap, and their sum is not 100%.

Five discovery source labels account for 9,743 of 11,068 observations (88.0%). Figure 7 includes the remaining labels in one aggregate bar. Source caps and collection failures can affect this mix. Of the 6,546 artifacts, 59 have observations from multiple sources. Missing or inconsistent identifiers may prevent additional cross-source matches.

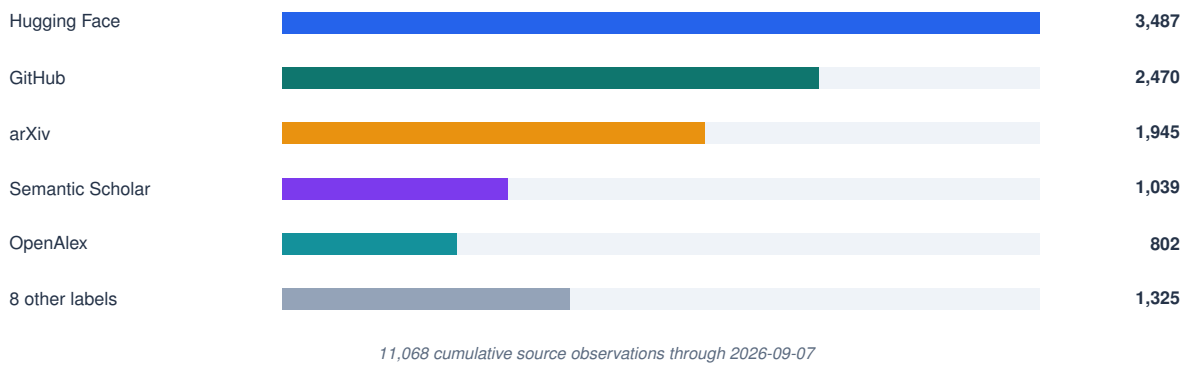


Figure 7: Discovery observations by source through 2026-09-07. Repeat sightings count as observations.

Figure 8 complements source coverage with the daily reading view. Its category cards separate new releases from updates, while the chart exposes changes in surfaced evidence and attention alongside collection failures.



Figure 8: Discovery trends through September 7, 2026. Category cards show newly surfaced releases and track updates separately; the bars show daily evidence and attention volume. Category tags overlap. The visible coverage notice reports failed sources. Click the image to open Trends.

5 Worked Example: Checking Prior Art

Before designing a new evaluation, a contributor surveyed August work on credit assignment in agentic training, with small Qwen-series models as a requirement for reproducible baselines. A coding agent installed the Benchmark Radar client and its public Skill, downloaded the corpus, and searched locally. It inspected the recorded paper, repository, and dataset links, tried additional web searches for work described in different terms, and read the source evidence before assembling the related-work table in Table 4.

The contributor used the table to assess whether the proposed evaluation duplicated existing work. The workflow separates candidate retrieval from comparison: Benchmark Radar retrieves candidates and exposes their evidence; the researcher or agent judges their relevance and compares the designs. Appendix E retains the session screenshots and an earlier manually assembled comparison.

#	Work	Published	Qwen base model(s)	Credit-assignment focus	Agentic benchmarks used	Link
1	SRPO: Self-Reflective Policy Optimization for Long-Horizon Reasoning	2026-08-25	Qwen3-8B (also Qwen3-1.7B / 32B scaling)	Reflection-conditioned dense token-level signals convert sparse terminal reward into training signal	AIME'24, WebShop, ALFWorld, SWE-Bench-Lite	https://arxiv.org/abs/2608.23493 https://github.com/Galleons2029/SRPO
2	ContextPilot: Teaching Agents for Proactive Context Management via Fine-grained RL	2026-08-28	Qwen3-8B, Qwen3-14B	Coarse-grained → action-level credit assignment over branched trajectories during agentic RL	InfBench (InfiniteBench), NovelQA, LongMemEval, BrowseComp+	https://arxiv.org/abs/2608.28476 https://github.com/Tencent/ContextPilot
3	SkillGate: Training In-Policy Skill Selection in Long-Horizon Agents	2026-08-21	Qwen3.5-9B (SFT checkpoint, RL init)	Selector credit starvation; partitions token support into two credit channels (outcome vs. action-local advantage)	5 agentic benchmarks incl. SkillsBench, SWE (16-candidate skill slate)	https://arxiv.org/abs/2608.18852
4	CIPO: Contextual Information Policy Optimization for Search Agents	2026-08-06	Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct	Dense turn-level credit (EALR) to evidence-using reasoning actions, combined with outcome reward	HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle + 3 OOD	https://arxiv.org/abs/2608.06128
5	MoRSE: Task-Oriented Multi-Agent System with Mixture of Role-Subtask Experts	2026-08-10	Qwen3-4B-Instruct (one of three backbones; also Llama-3.1-8B, Gemma-4-31B)	Hierarchical GRPO with two-layer credit assignment (isolates expert vs. routing quality)	Code-generation benchmarks (multi-agent), held-out task domains	https://arxiv.org/abs/2608.09251

Table 4: Summary table of recent work on credit assignment in agentic training, assembled during the session. Each row links to its arXiv record and, where available, its repository.

6 Worked Use Case: Selecting an Evaluation Suite for a Scientific Software Agent

We evaluated a concrete decision faced by a research engineer maintaining scientific Python repositories: which benchmark suite should be used for an agent that triages an issue, retrieves relevant papers and code, proposes a patch, and verifies the change? An incorrect patch or unsupported scientific claim can waste expert review time and invalidate downstream analysis. We predeclared two inclusion rules: a candidate had to exercise scientific research or repository-level engineering, and its task, metric, and evaluation protocol had to be inspectable from a cited source. General coding scores without a research or repository interpretation were excluded.

We ran the following searches against the frozen v0.11.0 catalog (software commit `8f46bbf`, discovery cutoff 2026-09-07): `PaperBench` and `software engineering agent`. We then inspected the three source records `model-reports:swe_bench_science`, `opencompass:1730`, and `llm-stats:swe_lancer` with `benchmark-radar show`. The machine-readable comparison, commands, snapshot, and record identifiers are committed in `docs/technical-report/agent-evaluation-use-case.yml`.

Candidate	Task fit	Protocol and evidence	Evidence gaps	Decision
SWE-bench Science [49]	High for scientific issue triage and patch verification	119 repository tasks across 98 repositories and 20 scientific domains; three task paradigms	Radar records the paper but no repository or dataset link	Primary suite
PaperBench [65]	High for literature-led replication and experiment execution	20 ICML 2024 paper-replication tasks decomposed into 8,316 rubric-graded subtasks with hierarchical judging	Paper and repository are recorded	Complementary research track
SWE-Lancer [44]	Medium for general software engineering	Freelance software tasks with client communication and real-world constraints	Radar records a repository but no paper or dataset link	External-validity control

Table 5: Evidence-backed evaluation-suite selection for a scientific software research agent. The systems are not combined into one score because their task units, verifiers, and protocols differ.

We therefore recommend a two-track primary suite: SWE-bench Science for repository repair and PaperBench for evidence-producing research replication. SWE-Lancer remains useful as a non-scientific engineering control, but it should not determine whether the research agent is suitable for scientific work. The decision is a recommendation from the stated workflow and source evidence, not a claim that either primary system is complete or that their scores are directly comparable.

7 Limitations and Future Work

The census describes this catalog at its recorded cutoff. Collection limits, failed requests, missing identifiers, and different snapshot dates affect its coverage. Source records are not a count of distinct

underlying tests. Broader coverage requires additional source collection and review of the evidence already present.

Retrieval precision, task suitability, and time saved remain to be evaluated. The worked example combines local queries with web search and has no controlled baseline. Lexical matching can miss paraphrases and renamed tasks. Evaluating semantic retrieval [53] will require reviewed relevance judgments across queries and candidate records, including less familiar benchmark families.

The arXiv discovery route collects new submissions; it does not backfill papers first posted before the collection window when later versions appear. The searches used for Section 2 missed two relevant earlier studies [1, 79], which a coauthor identified and read.

Measuring benchmark saturation or score stagnation requires comparable test versions and settings, with dates tied to score reporting or evaluation.

Task-capability classification also remains unvalidated in this rebuild. The deterministic null extractor used in CI assigns no capability levels to its 5,863 discovery-derived tracks. Completing and evaluating those labels is a separate task from maintaining the benchmark catalog.

8 Conclusion

Benchmark Radar brings daily benchmark discovery, catalog search, model-report mentions, and score histories into one living search engine. Readers can find candidate evaluations, locate task materials, inspect reporting choices, and follow the settings and citations behind a score. Shared benchmark IDs connect the web dashboard, downloadable catalog, and offline query clients.

The v0.11.0 release makes 1,283 source records searchable and preserves 464 unscored records with artifact links. Its full-catalog analyses distinguish benchmark records, scored models, cited documents, and numeric observations.

Author Contributions

Koutian Wu led the work and manuscript preparation, built the initial collection pipeline, and implemented data aggregation across sources. Junjie Zhou prepared the score-archive audit and contributed report revisions. Wenjun Zhang contributed discussion and improvements to Benchmark Radar. Ergan Shang prepared figures and contributed copyediting. Jiayu Wang contributed the worked use case and its supporting evidence. Pengqian Han contributed data analysis and copyediting. Junkai Wang and Wanghan Xu contributed review and copyediting. Songyuanyi Lu contributed discussion and improvements to Benchmark Radar. Rudy Mizrahi Celekli, Lazizbek Ravshanov, and Yaoliang Bian contributed to software development for Benchmark Radar. Lin Shi contributed to conceptualization and methodology, and advised the work.

Acknowledgements

We thank Ruohan Xu, Yifan Xu, Ran Hong, Hejia Geng, Haozhe Zhang, and Zhenxin Huang for discussion and for improving Benchmark Radar.

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Marek Šuppa, Jan Batzner, Jenny Chim, Jeba Sania, Yanan Long, Hossein A. Rahmani, Christina Knight, Yiyang Nan, Jyoutir Raj, Yu Fan, Shubham Singh, Subramanyam Sahoo, Eliya Habba, Usman Gohar, Siddhesh Pawar, Robert Scholz, Arjun Subramonian, Jingwei Ni, Mykel Kochenderfer, Sanmi Koyejo, Mrinmaya Sachan, Stella Biderman, Zeerak Talat, Avijit Ghosh, and Irene Solaiman. When AI benchmarks plateau: A systematic study of benchmark saturation, 2026. URL https://arxiv.org/abs/2602.16763. Published at ICML 2026.

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Appendix

A Full-Catalog Census

This census retains all 1,283 source records, including those without scores.

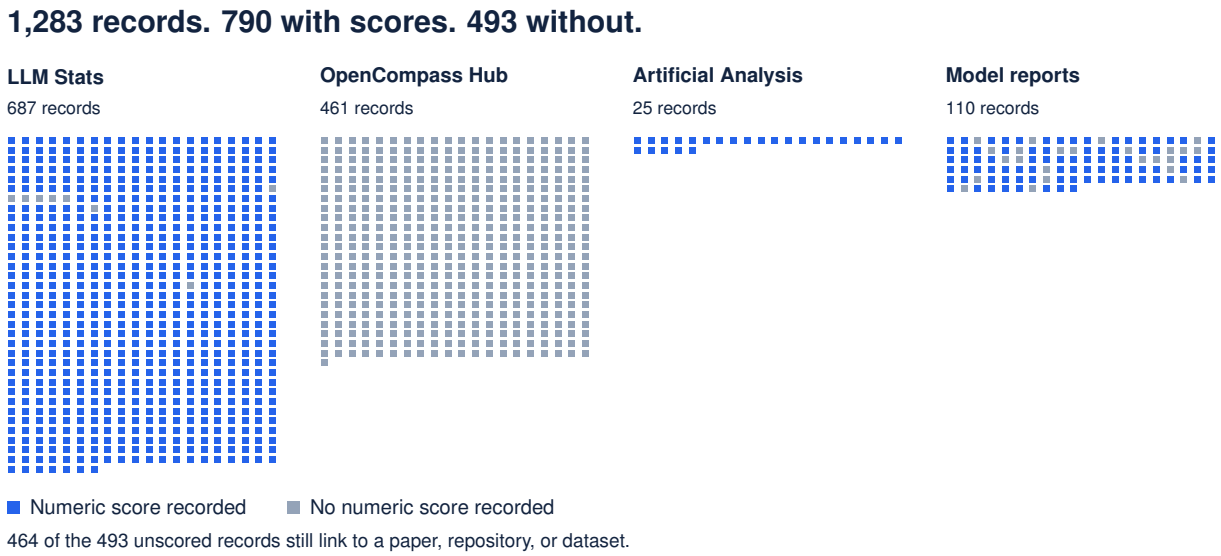


Figure 9: 464 unscored entries provide artifact links. Each mark represents one source record and links to its detail page. All 1,283 records appear, without a date or score filter.

B Catalog Evidence and Measurement Coverage

The following analyses use the same catalog as Section 4. They document the evidence behind its records and the requirements for comparing measurements.

B.1 Cited Documents and Scored Models

The common document registry contains 1,208 distinct cited documents, attached to 1,278 of the 1,283 benchmark records. 5 records have no cited documents. The shared model registry contains 868 model identities. Model identifiers are available for every numeric score in all 790 scored records, allowing distinct models to be counted within each record. For unscored records, this count is unknown.

Table 6 illustrates why these counts cannot substitute for one another. The Artificial Analysis record for GPQA Diamond contains scores for 586 models and cites 1 registry page. The Model reports record cites 27 documents and contains 21 numeric observations for 19 models.

Benchmark record	Source	Models	Documents	Scores
GPQA Diamond	Artificial Analysis	586	1	586
GPQA Diamond	Model reports	19	27	21
Humanity's Last Exam	Artificial Analysis	577	1	577
SciCode	Artificial Analysis	577	1	577
CritPt	Artificial Analysis	492	1	492

Table 6: One cited page can document hundreds of scored models. Models count distinct source model IDs within a record; documents count distinct citation IDs; scores count numeric observations. These selected examples illustrate the units. The census retains all source records.

Table 6 also records 577 scored models for Humanity's Last Exam, 577 for SciCode, and 492 for CritPt, each under one Artificial Analysis citation.

B.2 Score Scales and Comparison Eligibility

Only 82 of the 790 scored records declare a percentage unit with a known direction and values within 0–100. The remaining 708 scored records use other or unverified scales. We retain their numeric observations, which do not support a shared percentage-headroom calculation. Table 7 accounts for the entire catalog before any such comparison.

Measurement state	Records	Interpretation
Numeric scores, declared percentage scale	82	Supports scale-specific summaries; protocols still need checking.
Numeric scores, other or unverified scale	708	Retain scores; do not assume a 100-point ceiling.
No numeric score	493	Score headroom remains unknown.
Full population	1,283	Every source record remains in the census.

Table 7: Only 82 of 790 scored records meet the percentage-scale rule in Section 3.5. All 1,283 records remain accounted for.

The catalog records benchmark release dates for 615 records, leaving 668 without a release date. Across score observations, 12,594 entries carry model-announcement dates and 322 carry document-publication dates. Neither date basis directly establishes an evaluation date. Researchers must first establish dates and comparable evaluation settings for the relevant source records.

A small gap to a metric ceiling can describe a reported setup. Catalog-wide headroom or recency estimates require the eligible measurement coverage alongside the statistic.

B.3 Documentation Across All Catalog Sources

The documentation analysis covers all 1,283 benchmark records. It finds 1,208 distinct cited documents supporting 1,278 records; 5 records have no cited document. Table 8 includes every source and keeps benchmark records separate from document counts. A registry page and a model report both count as one cited document when attached to a record. Repeated citations to the same document do not increase its count.

Source	Records	With docs	Documents	Docs without named org.
LLM Stats	687	687	687	687
OpenCompass Hub	461	461	461	461
Artificial Analysis	25	25	23	23
Model reports	110	105	37	0
Total	1,283	1,278	1,208	1,171

Table 8: Documentation across the complete v0.11.0 catalog. "With docs" counts benchmark records; the last two columns count distinct documents. Missing organization metadata remains unknown.

Documentation is widespread, but the release data does not identify an organization for 1,171 cited documents. The complete export retains each document's identity and source URL on its benchmark record.

B.4 Reported Scores Across the Full Catalog

The score analysis retains all 1,283 records and their 12,916 numeric observations. Table 9 summarizes scored-model coverage for each source. 22 benchmark records contain numeric scores for at least 100 distinct source model IDs. The 708 records on other or unverified scales contribute their scores and model counts on the same terms as percentage-scale records. The 493 records without numeric scores remain in the population with unknown scored-model coverage.

Source	Records	Scored	Unscored	Score rows	Median models	Max. models
LLM Stats	687	679	8	5,544	3	239
OpenCompass Hub	461	0	461	0	Unknown	Unknown
Artificial Analysis	25	25	0	7,050	245	586
Model reports	110	86	24	322	2	19
Total	1,283	790	493	12,916		

Table 9: Reported scores across every source in the v0.11.0 release. Models count distinct source model IDs within each benchmark record. Medians and maxima use records with known model counts; unscored records are not assigned zero models. Model counts are not summed across benchmarks. The record-level score export contains every catalog record, including raw maxima, all tied reporting models and citations, units, settings, and explicit missing values.

The accompanying `evidence/catalog-findings.csv` contains one row per benchmark record, with a JSON counterpart retaining detailed evidence. Each scored record has its highest numeric value on the source's native scale and every tied score observation. Units and score direction remain attached

to that record; a numeric maximum need not be the best result for a lower-is-better metric. Display multipliers are recorded separately and do not establish a percentage unit.

Percentage headroom remains a separate calculation: 100 minus a record's highest value, only when its unit is explicitly percent, its direction is higher-is-better, and its values stay within 0–100. Missing scale evidence leaves headroom unknown while preserving the scores themselves. The export records eligibility for every benchmark. Test versions, reasoning budgets, tools, attempts, and evaluators must still be checked before comparing results [14, 43].

B.5 Date Evidence Across the Full Catalog

The date analysis also retains all 1,283 source records. Table 10 separates benchmark release dates from dates attached to numeric observations. A known model announcement date supplies a source-record proxy, not an evaluation date.

Source	Records	Release known	Doc.-dated scores	Model-dated scores	Undated scores
LLM Stats	687	42	0	5,544	0
OpenCompass Hub	461	457	0	0	0
Artificial Analysis	25	17	0	7,050	0
Model reports	110	99	322	0	0
Total	1,283	615	322	12,594	0

Table 10: Date evidence for the complete v0.11.0 catalog. The first two numeric columns count benchmark records; the final three count numeric score observations. Records with no scores remain in the Records column.

The exported rows preserve each record's release date and the date basis of its scores. A study of score progress needs actual reporting or evaluation dates and comparable settings for the records being studied.

C Collection Settings and Display Filters

The 2026-09-07 snapshot records 1,003 fetched rows and 954 candidates after duplicate removal. Of these, 366 qualified for publication and 138 met the recommendation threshold. These are discovery-run counts, separate from the benchmark catalog census. Published evidence retains source URLs, retrieval times, parser versions, identifiers, and payload checksums. Raw responses and credentials remain outside the public artifact.

Benchmark Frontier, on the Leaderboard tab, requires a numeric score and excludes known pre-2024 benchmarks. It retains scored records with unknown dates or counts through labelled marks, and keeps unverified scales outside Pareto calculations. Its initial score cutoff is 70. The linked score ranking retains its date and numeric-score requirements independently of that cutoff. General catalog search and exports retain the full population. The census in this paper applies none of these display filters.

A separate priority score for daily recommendations combines relevance (35%), evidence (20%), recency (20%), and adoption (25%) on a 0–100 scale. These weights describe reading priority for discovery observations. An agent assessing benchmark suitability can try focused query variants, inspect record details, and read the cited sources; that judgment occurs after candidate retrieval.

D Source Inventory and Health

Benchmark Radar ingests from 37 sources: 13 direct connectors and 24 first-party feeds. The direct connectors read arXiv [6], Hugging Face Hub [24], GitHub Search and GitHub Releases [18, 19], Kaggle [28], Zenodo [80], Crossref [12], OpenAlex [52], OpenReview [50], Semantic Scholar [31], Brave Search [8], and Hacker News [2]. Table 11 lists source connectors that completed successfully at the cutoff. Row counts are measured before duplicate removal and ranking. A connector can be healthy with zero eligible rows if its request succeeds and the response can be parsed. Core sources cover papers, shared model or dataset artifacts, and code; all must be healthy before publication. Failures in optional sources are reported but do not block publication.

Connector	**Role**	**Cutoff status**	**Core**
arXiv	Primary papers via selected AI, language, vision, and software categories	Healthy; 46 rows	Yes
Hugging Face Hub	Datasets and Spaces connected to benchmark artifacts	Healthy; 101 rows	Yes
GitHub Search	Code repositories and benchmark artifacts	Healthy; 300 rows; at cap	Yes
GitHub Organizations	Reviewed organization repositories	Healthy; 6 rows	No
Hugging Face Papers	Community-surfaced papers	Healthy; 26 rows	No
Kaggle Datasets	Public benchmark datasets	Healthy; 30 rows	No
Zenodo	DOI-bearing research artifacts	Healthy; 65 rows	No
Crossref	DOI metadata	Healthy; 180 rows	No
OpenReview	Conference submissions	Healthy; no eligible rows	No
GitHub Releases	First-party releases	Healthy; 1 row	No
OpenAlex	Scholarly discovery metadata	Healthy; 248 rows	No

Table 11: Healthy direct discovery connectors at the current cutoff.

Table 12 lists research and engineering feeds operated by the institutions being monitored. They can surface benchmark papers, datasets, model announcements, and engineering posts. The feed collector returned no records at the cutoff. Semantic Scholar returned a malformed payload, and Brave Search lacked an API key. These optional-route failures remain recorded in the snapshot. Searches restricted to an organization's website cover institutions that lack a verified feed.

Feeds 1-12	Feeds 13-24
Meituan Engineering	Ai2
OpenAI News	Together AI
Google AI	Sakana AI
Google DeepMind	Qwen
Google Research	Ollama
Apple Machine Learning Research	Stability AI
AWS Machine Learning	Nomic AI
Hugging Face Blog	Replicate
Microsoft Research	NVIDIA Developer
NVIDIA AI Blog	IBM Research
Mistral AI	Databricks
Meta Research	LangChain

Table 12: The 24 first-party research and engineering feeds monitored by Benchmark Radar.

E Worked Use Case: Prior-Art Check for a New Evaluation

The contributor wanted to determine whether a proposed evaluation duplicated existing work. They surveyed work published in August 2026 on credit assignment in agentic training, using small Qwen-series models as a requirement for reproducible baseline experiments. The search produced a focused comparison table that the contributor used to assess the proposed design.

The contributor gave the task to a coding agent together with the public setup instructions for Benchmark Radar. The agent installed the command-line client and the Benchmark Radar Skill, a set of instructions for using the client. It then downloaded the local corpus and queried candidate records offline (Figure 10).

The agent inspected paper, repository, and dataset links to decide which sources to open next. Figures 11 and 12 show two retrieved discovery records: one has a repository link, and the other has both paper and repository links. Neither records a dataset link, leaving dataset availability for a follow-up check.

The agent supplemented local queries with web search to find related work described in different terms. It cross-checked the candidate sets and inspected the source evidence before selecting records for the comparison. Figure 13 shows the word matches and score components available for inspecting one retrieved candidate.

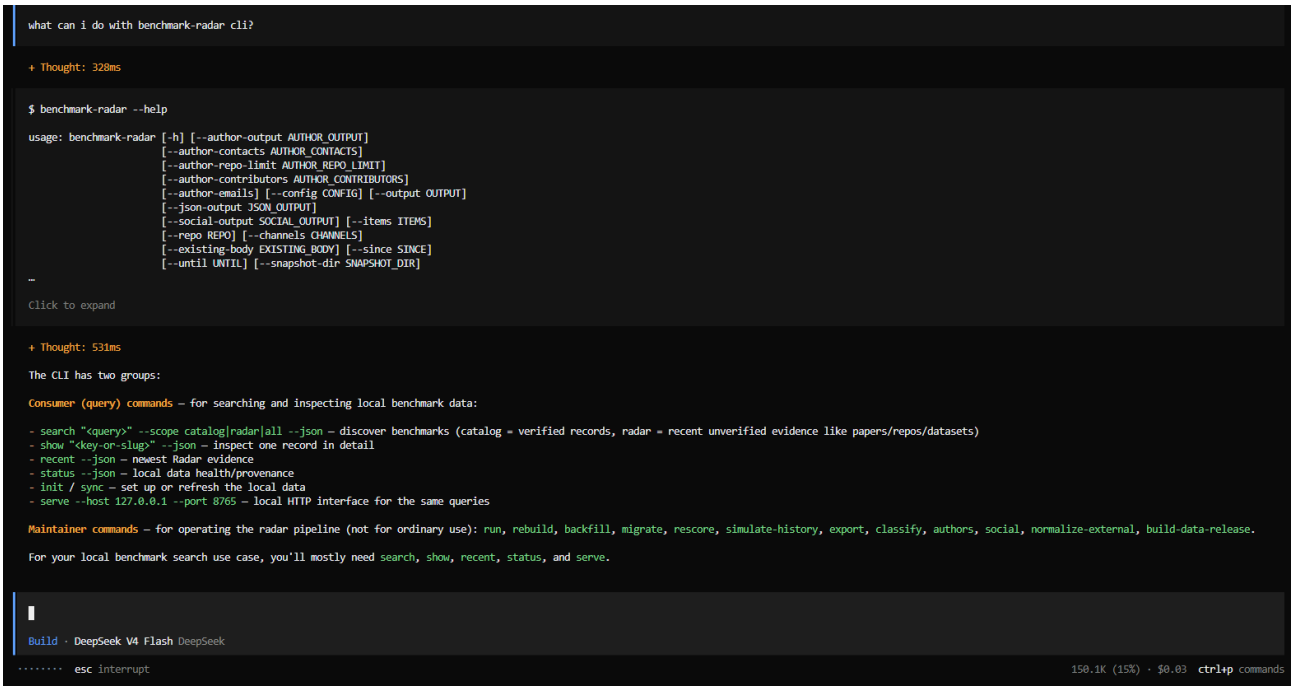


Figure 10: A coding agent installs the Benchmark Radar client and its usage instructions, then runs local queries.

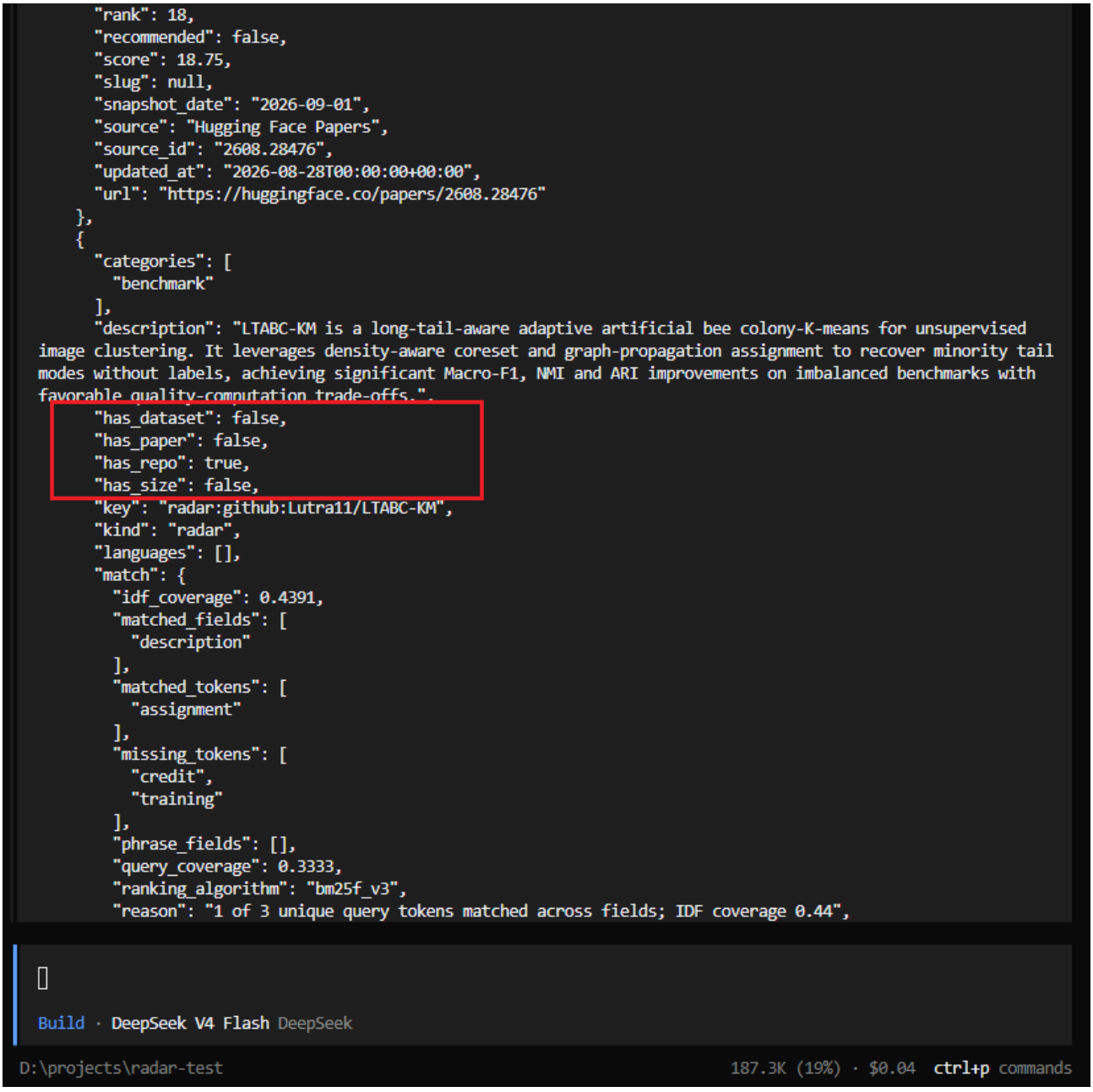


Figure 11: A retrieved discovery record with a repository link for inspecting the implementation. Paper and dataset links are missing from the record.

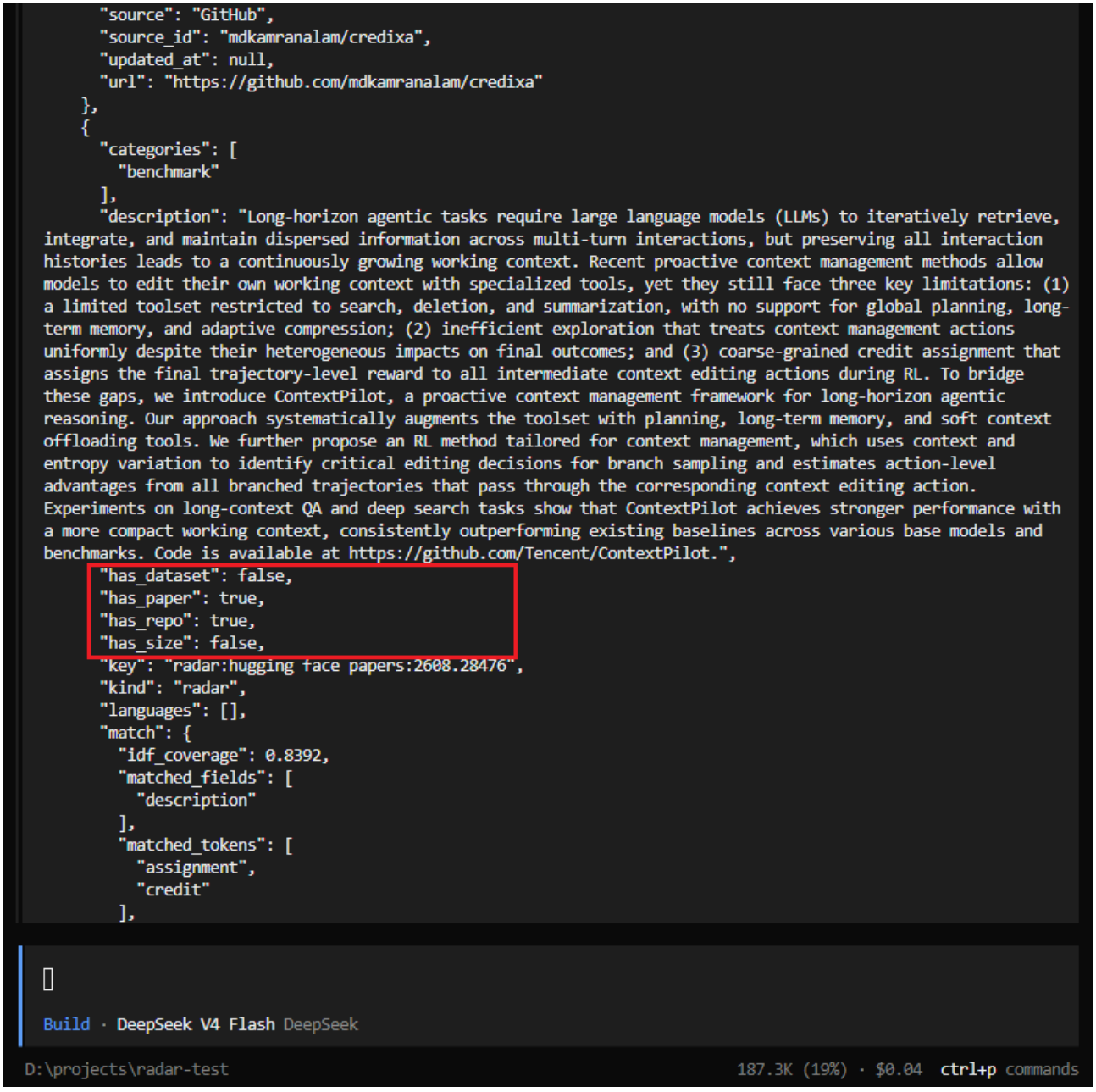


Figure 12: A second discovery record provides both paper and repository links. Its dataset link needs a follow-up check.

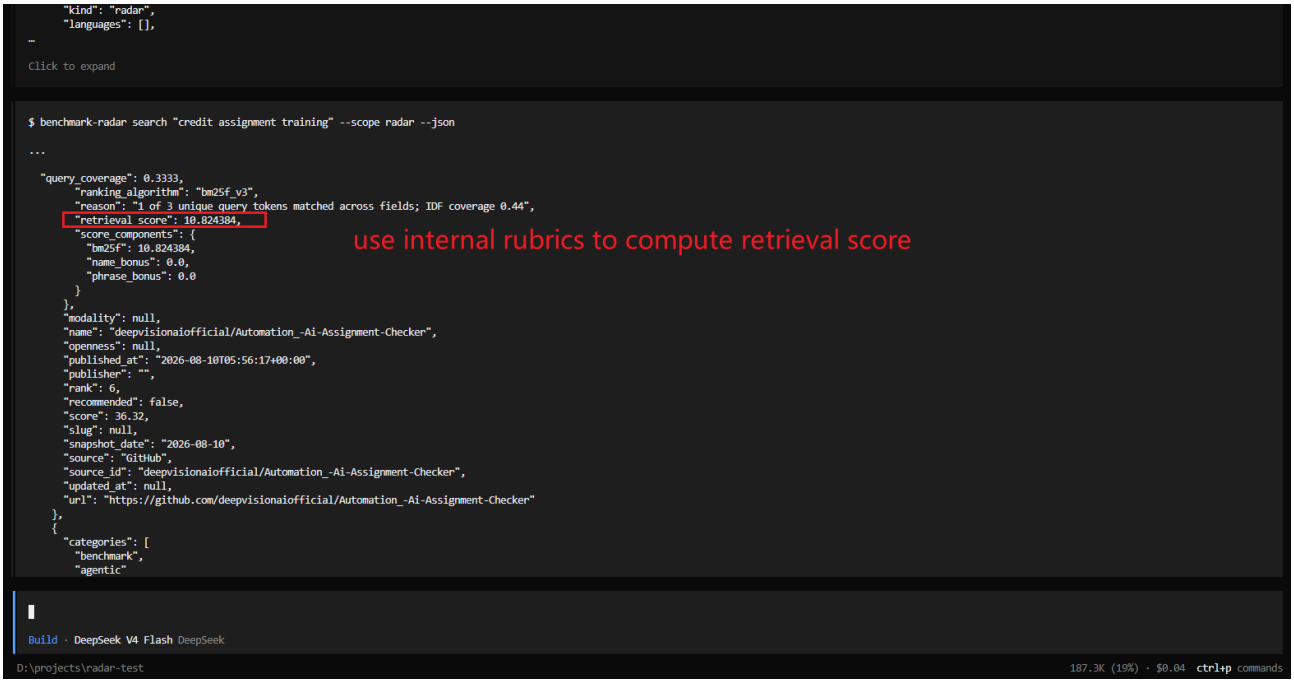


Figure 13: A candidate record with query-word matches and retrieval-score components for the agent to inspect when judging relevance.

The resulting related-work table appears in Table 4 in the main text.

Table 13 shows a prior-art comparison table from an earlier benchmark-design effort by one contributor. Each row is one related benchmark and each column one design dimension, with each cell read from a source paper. The contributor reported that assembling it took more effort than any part of that project except producing the benchmark data. Benchmark Radar retrieves candidates for this kind of comparison and exposes the matching words and fields for inspection; the comparison itself requires reading the source evidence.

Bench Name	End-to-End Tasks	Fine-Grained Eval	Researcher-Quality Eval	Data Generation	Multi-Harness Eval	#Tasks
MLE-Bench [9]	✗	✓	✗	Transfer&Compose	✓	75
MLGym-Bench [45]	✗	✓	✗	Automatic	✗	13
EXP-Bench [32]	✓	✗	✗	Automatic	✓	461
ResearchCodeBench [23]	✗	✓	✗	Transfer&Compose	✗	212
MLR-Bench [10]	✓	✗	✗	Automatic	✓	201
PaperBench [64]	✗	✓	✗	Transfer&Compose	✗	8316
AstaBench [7]	✓	✓	✗	Transfer&Compose	✓	2400+
InnovatorBench [75]	✓	✗	✗	Transfer&Compose	✗	20
AIRS-Bench [40]	✗	✓	✗	Automatic	✓	20
COMPOSITE-Stem [71]	✓	✓	✗	Manual	✗	70
ScienceBoard [67]	✗	✓	✗	Manual	✗	169

Table 13: A prior-art comparison table assembled manually for an earlier benchmark-design effort. Each row is one related benchmark and each column one design dimension, all read from the source papers by hand.

F Reproducibility, Access, and Citation

This revision uses the Benchmark Radar v0.11.0 release, software commit 8f46bbf, with a discovery cutoff of 2026-09-07. The release preserves the inputs already audited for this paper; later data is outside its scope. A fresh detached checkout passed the six required CI steps in order: linting, formatting checks, catalog normalization, classification, release construction, and tests. The dated registry inputs remain the August snapshots registered in `data/leaderboard_snapshots.yml`; the September cutoff does not imply that those sources were recrawled on that day.

The data build normalizes the registry and model-report inputs into the shared catalog, then classifies discovery tracks and packages the catalog with checksums. Installed clients validate an update before activating it. The manuscript and its dated analysis files build independently of the software checkout. The project provides a DOI and Citation File Format record for citation [11, 30, 63, 72]. Table 14 lists the access and citation resources.

Artifact	Canonical or permanent location
Archived paper (v0.9.0)	https://doi.org/10.5281/zenodo.22167102
Release data (v0.11.0)	https://github.com/ktwu01/benchmark-radar/releases/tag/v0.11.0
Source code	https://github.com/ktwu01/benchmark-radar
Dashboard	https://benchmark-radar.org/
Discovery observations (JSON)	https://benchmark-radar.org/data/radar.json
Benchmark catalog	https://benchmark-radar.org/data/benchmark-index.json
RSS	https://benchmark-radar.org/feed.xml
Citation metadata	https://github.com/ktwu01/benchmark-radar/blob/main/CITATION.cff

Table 14: Access and citation artifacts for Benchmark Radar.

The census reads `site/data/benchmark-index.json`, the detail shards under `site/data/benchmarks/`, and the shared document and model registries. Discovery counts come from the rebuilt `site/data/radar.json` and dated snapshots. Model reports and score YAML files are normalization inputs, not a separate population for the paper's findings.

The paper repository contains `scripts/audit_catalog.py`, which validates IDs and counts, computes the census, and exports `catalog-data.tex` plus `evidence/catalog-audit.json`. The JSON includes every source-record key, measurement state, per-record model and document counts, citation IDs, and SHA-256 hashes of the inputs. The script checks score-observation uniqueness, record-to-shard identity, document-registry coverage, and reconciliation of scored and unscored records. Its `--check` mode compares a fresh calculation against the committed outputs. Figure 9 draws one linked dot per census record in source-key order.

Reproduce from a clean checkout at software commit 8f46bbf using the README's six-step CI sequence. The software exporter writes `figure-data.tex`; the paper's `audit_catalog.py` and `audit_findings.py` take the rebuilt software path. The findings audit writes `findings-data.tex` and `evidence/catalog-findings.json` plus a CSV export. Every analysis retains exactly the catalog's source-record IDs. It records native score maxima and all tied observations, units, settings, cited documents, model counts, and date evidence. Table cells and repeated numeric claims are generated from these same inputs. Run each exporter's `--check` mode, build with `make arxiv`, compile the extracted package, and inspect the PDF. Normal builds use the committed inputs. The v0.9.0 DOI remains an earlier deposit; live dashboard counts require a retrieval date.

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